

Contemporary C++: *Learning Modern C++ in a Modern Way*



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Some fundamental abstractions: Constants, Pointers, and References

Part 1

- Constants
- Pointers
- Null pointer literal: `nullptr`
- References
- References and further readings
- Q & A

120 min (incl. Q & A)



Constants

- The keyword `const` can be added to the declaration of an object to make the object declared a constant.
- Constants must be initialized.
- Constants can't be assigned.

Symbolic
variable

→
`const`

Symbolic
constant

```
const float PI = 3.14; // PI is r-value
const int light_speed = 3 * 100000000; // m/s
const double cm_per_inch; // error: no initializer
PI++; // ++ needs lvalue
const char vowel[5] = { 'A', 'E', 'I', 'O', 'U' };
```

- (Obviously) you can assign a const to a non-const but not a non-const to a const.
- Constants → maintainable code
- Constants → more localized information
- Use symbolic constants to avoid *magic numbers*.

```
const int max = 128;
int v[max];
```

```
for (int i = 0; i < BUF_SIZE; i++)
    // ...
```



Multi Centimeter Inch Conversion

Prog.



Centimeter Inch Conversion 2

Prog.

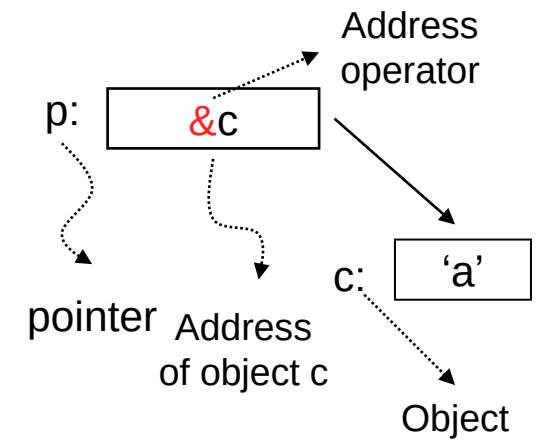
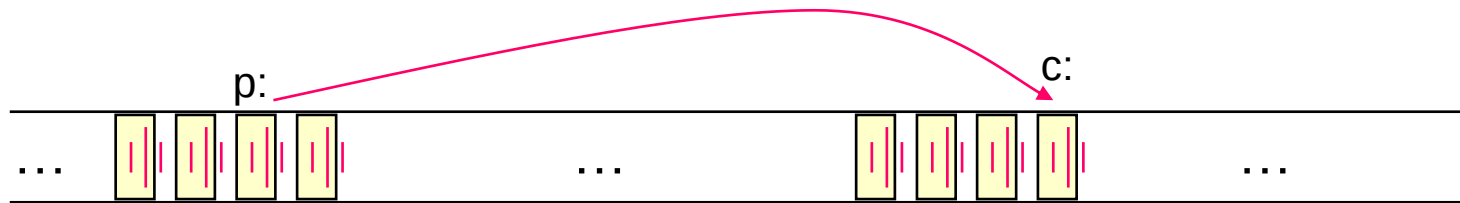
Pointers: An introduction

- A typical machine has an array of consecutively numbered or addressed memory cells.



- For a type T , T^* is the type “pointer to T .” That is, a variable of type T^* can hold the address of an object of type T .
- *Pointer* is an object whose value represents the address of another object or 0.
- If you can access an object, you can obtain its address, and vice versa.

```
char c = 'a';  
char *p = &c; // p holds the address of c
```



Pointers: cont.

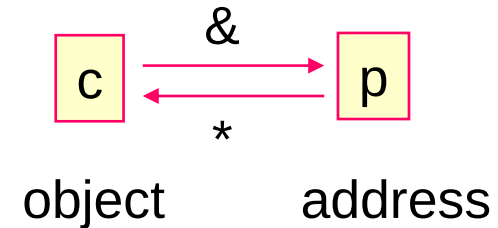
- The fundamental operation on a pointer is *dereferencing*, that is, referring to the object pointed to by the pointer. This operation is also called *indirection*.

```
char c2 = *p; // c2 == 'a'
```

Dereference
operator

- Indirection operators:

Operator	Function	Use
&	address of	&Ivalue
*	dereference	*expr



R references and further readings

[KR78] Brian W. Kernighan & Dennis M. Ritchie. The C Programming Language, Prentice-Hall, Englewood Cliffs, NJ, 1978.

[Str00] Bjarne Stroustrup. The C++ Programming Language (Special Edition). Addison Wesley. Reading Mass. USA. February 2000.

[Str08] Bjarne Stroustrup. Programming: Principles and Practice in C++, 1st edition, Addison Wesley, December 2008.

Thanks for your patience ...

A man who asks a question is a fool for minute,
The man who does not ask, is a fool for a life.
- Confucius

Learning to ask the right (often hard) questions is an essential part of learning to think as a programmer.

- Bjarne Stroustrup *programming Principles and Practice Using C++, page 4.*

There is no stupid question, but there is stupid answer.
- Howard Hinnant

